

SENSEI

AI-Powered Navigation and Accessibility Companion

Final Year Project Proposal

Session 2022-2026

A 3rd Year Student

A project submitted in partial fulfilment of the
COMSATS University Degree
Of
BSc. (Hons.)BS in Software Engineering (CUI)



Department of Computer Science
COMSATS University Islamabad, Lahore Campus

11 March 2025

Project Detail

Type (Nature of project)	☒ Development ☐ Research ☐ R&D			
Area of specialization	Mobile App Development			
Project Group Members				
Sr. #	Reg. #	Student Name	Email ID	*Signature
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*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others

Plagiarism Free Certificate

This is to certify that, I am Jaisha Sohail S/D/o Malik Sohail Jamal, group leader of FYP under registration no CIIT/FA22-BSE-207/LHR at Computer Science Department, COMSATS University Islamabad, Lahore Campus. I declare that my FYP proposal is checked by my supervisor and the similarity index is _____% that is less than 20%, an acceptable limit by HEC. Report is attached herewith as Appendix A.

Date: _____ Name of Group Leader: Jaisha Sohail

Signature: _____



Name of Supervisor: Dr. Tariq Umer

Co-Supervisor (if any): _____

Designation: _____ Associate Professor _____

Designation: _____

Signature: _____



Signature: _____

Abstract

This document outlines the development of “Sensei” , an app that helps blind people navigate with AR. It gives real time information about objects, obstacles and people with sound, vibrations and AI powered object recognition.

Unlike simple tools that only give voice commands, Sensei gives real time guidance, detects emotions during conversations, reads text and lets you customize sounds for better navigation. It makes moving around familiar places – home, streets and public areas – easier and safer.

Development process is step by step from planning to testing to launch. The project is divided into phases: conceptualization, design, development, testing, and deployment, with progress tracked using tools like Gantt Charts.

This report basically provides an overview of the problem statement , objectives and requirements including functional and non-functional . It concludes with a discussion of the challenges faced , lessons learned along with future enhancement areas such as multi-language support and advanced AI integrations .

Sensei is more than a tool – it’s a smart companion that helps visually impaired people to move freely, interact with others and live more independently

1 Introduction

“Sensei” is a mobile application that is built solely for the purpose of helping visually impaired users with their smooth walking . With constant navigation done with the help of Augmented Reality (AR) , this app stands out for its continuous haptic AI feedback and audio commands . Its basic goal is the independence of the blind user . From basic route navigation to social interaction , this app helps with it all so that the blind user doesn’t lack in any way and becomes part of the society , almost as if they are not visually impaired at all . This built-in AI instructor will act like a real human guiding companion , like a real friend , to keep the user engaged and make their daily life easier and almost exciting .

The name “Sensei” is actually a Japanese term meaning “teacher” or “Guide” , helping users navigate their environment and guide them throughout their journey . Whether it's a trip to the grocery store or a trip to the mountains , “Sensei” handles it all ! All the user has to do is open the camera in the app through voice commands and AR devices will be able to guide the user in real-time through continuous sensory feedback .

With advanced tools and technologies , this app will use algorithms to help with the foot placement , object detection and emergency alert systems to ensure the safety of the user . It also includes special features like 3D spatial audio and personalized route memory . This app will use AR functionality with personalized AI , AI frameworks for object detection and API’s for real-time data processing .

In short , “Sensei” aims to redefine modern assistive technologies by combining cutting-edge AR capabilities with user-centric design offering a unique solution to the blind community .

1.1 Objectives:

Sensei's main goals are described as follows:

- Improving the awareness of the environment
- Improving navigation and mobility
- facilitating social interaction
- providing contextual information
- Integrating intelligent features
- ensuring safety and support for emergency rooms
- Promoting personalization and learning
- Foster independence and confidence

1.2 Problem Statement:

Modern technological devices feature restricted operational capabilities through their use of standard voice direction with automated navigation that fails to maintain dependable assistance. Sensei goes beyond traditional assistive devices through its flexible capabilities which include emotional detection for human social interactions as well as text reading from context and non-context sources and customizable audio controls.

1.3 Assumptions & Constraints:

1.3.1 Assumptions of Users

1.Familiarity with Mobile Devices:

- One presumption would be the user's ability to handle the smartphone and its basic features such as tapping, swiping and issuing voice commands .
- Users should understand the correct methods to wear and operate their haptic feedback devices including smart belts and wristbands.

2.Environmental Awareness:

- The system requires users to know their environment sufficiently to follow the audio prompts needed for navigating.
- The system requires users to utilize features that include obstacle detection and route memory as well as object recognition functions to improve their self-sufficiency.

3.Comfort with Audio Feedback:

- Users need to depend on 3D spatial audio and text-to-speech (TTS) outputs for current guidance as well as informational updates.

- The application requires users to change audio profile preferences which will guide their adjustment process.

1.3.2 App Control Inputs

1. Voice Commands

Users can give voice commands . Just by saying things like “Where is the door?” or “What am I holding in my hand ?” . The app listens and responds, using speech-to-text technology to understand commands.

2. Gesture Controls

For blind users who prefer touch-based navigation, the app supports simple gestures like swiping left or right to move menus, and tapping to select options. These gestures are easy, making the user's navigation smooth.

3. Haptic Feedback

The app connects with smart devices like belts or smartwatches that vibrate to guide users. These vibrations help detect and alert for obstacles in front , provide direction, and make movement safer .

With a mix of these , Sensei makes navigation effortless for everyone.

1.3.3 App Usage Assumptions

1. Engagement with Features – Users are expected to use app features given like obstacle detection, memorizing the overly taken route by the user and object recognition. The app assumes that users will rely on AI-powered descriptions (e.g. "This is a stop sign") and several emergency alerts for the safety of users.
2. Learning & Customization – Users are expected to open the learning mode to gain knowledge of their surroundings when new routes are taken or new environments are discovered . The app assumes users will customize their settings like audio profiles to meet their individual needs.
3. Smart Home Connection – Users are expected to connect the app with smart home devices to improve one’s control.

1.3.4 Constraints

The technical constraints, time constraints and the resource constraints within which the development team will function. Due to resource constraints , the development team will use Expo AR for Augmented reality function and AI frameworks for object recognition function for efficient resource utilization . Testing will be conducted on real-world scenarios with blind users to ensure whether the app is useful to them or not and could work in the market . The app will be required to work on smartphones with minimum system requirements for broad accessibility .

1.4 Project Scope:

“Sensei” is an ambitious project that combines modern and cutting-edge technology with an appealing design to create an augmented reality app for the blind community . The aim of this app is mainly to make the blind user independent of anything and be able to walk on their own without the help of anything or anyone . As long as the user has a smartphone in their hand , with voice control the user is able to access the app and navigate their way to any destination they want through assistive haptic and audio feedback .This also serves as a social interaction tool so that the user does not feel left out in any sitting and understands the environmental contexts .

At its core , “Sensei” focuses on innovation , making use of AR , AI and haptic feedback to create a seamless user experience . The core features of this app are , real-time navigation , Object detection , foot placement strategy , object recognition and emotion detection for any social interaction . These elements are the key features for the walking of a blind person .

“Sensei” is particularly special for its adaptive and dynamic functionality . The app uses AI-powered object recognition strategy and guides users through any harmful objects to ensure the safety of the user , even if the user has to go through an obstacle course. The app also mentions the danger level of the obstacle ahead of them to make the user extra aware and extra careful . It also contains emergency alert systems. If the user is too close to hitting something or someone or if any danger is approaching , the app through integration with smart wearables sends a vibrating signal to extra alert the user before any mishap occurs . The user can also change the app’s audio profile settings to suit their individual preferences for a more personalized experience .

Another one of the main features include foot placement strategy . With height and depth calculating algorithms , The app will guide the user to place their foot in a particular way to go through the obstacle such as any step or stairs . “Sensei” also specializes in guiding users through any difficult social interaction .

The app offers offline functionality for the main features . It will also use cloud-based backend systems for real-time data processing . Testing will be done with the help of blind users’ feedback to ensure that the walking is smooth and seamless with no safety concerns so that the user totally depends on this smart app .

Finally , “Sensei” will be designed to work on both iOS and Android for broad accessibility . The app’s user interface will prioritize simplicity and intuitiveness with features like gesture-based controls and voice commands making it easy to use .By combining innovative technology with a user-centric approach, Sensei aims to redefine assistive tools and empower visually impaired individuals to live more independently.

1.5 App Experience:

Sensei is not just a navigation tool, it's a personalized and adaptive companion that grows with the user. The app combines AR, AI, and audio as well as haptic feedback to create an experience that feels like having a trusted guide by your side. As the user is navigating the area, Sensei provides contextual information to make the experience more engaging. This feature makes the navigation into an educational and enjoyable experience.

Sensei provides a virtual companion to guide the user while having a unique personality. Users can choose from different companion styles. The companion adds a little bit of personalization and emotional connection, making the app more engaging. The app's learning mode is adaptive to the user's independent navigation progress, providing more or less guidance as needed. It provides cultural as well as historical information about landmarks, buildings, and public spaces and includes advanced safety features to protect users in emergencies. This provides peace of mind and ensures user safety in critical situations.

1.6 App Interface:

"Sensei" has a non-visual interface. The user only interacts with the app through audio commands and simple gestures and receives feedback through voice or vibrations so that it focuses solely on the smooth and safe navigation of the user. With the help of these functions, the app helps users to actually see their environment with only sound and touch without needing to rely on the screen but just their voice and natural hand gestures.

1.7 About Sensei:

"Sensei" is an augmented reality navigation app to empower visually impaired individuals. Here's some basic information about the app:

- **Platform:** iOS and Android (initial release), with potential for expansion to AR glasses and wearables.
- **Genre:** Assistive Technology, Navigation, and Accessibility.
- **App Style:** AR-based, audio-driven, and mission-oriented experience.
- **Perspective:** Audio-first interface with haptic feedback and voice guidance.
- **Setting:** Real-world environments (e.g., streets, homes, public spaces) transformed into accessible and navigable spaces.
- **Experience Type:** Adaptive and customizable, with mission-based objectives (e.g., finding a location, avoiding obstacles).
- **Target Audience:** Visually impaired individuals seeking independence and accessibility in their daily lives.

1.8 Tools & Technologies:

The following tools will be used for the development of Sensei:

- **Development Framework:** React Native with Expo.
- **AR Frameworks:** Expo AR (with Three.js) for AR functionality .
- **AI Integration:** TensorFlow Lite / PyTorch.
- **Audio Tools:** Expo AV and Google Resonance Audio / Apple Spatial Audio .
- **Wearable Integration:** Expo's Bluetooth API .
- **Programming Language:** JavaScript .

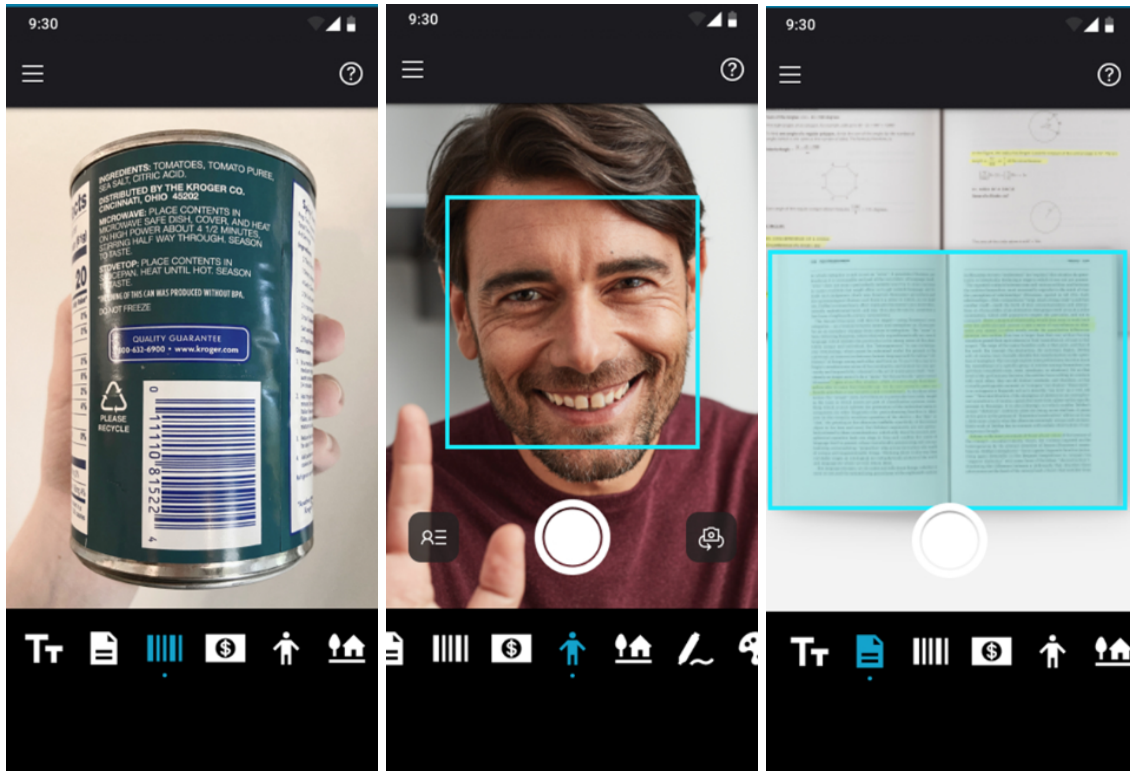
2 Related Work:

2.1 Literature Review:

Some similar games related to the proposed game are "Vision", "Look Out" and "Sullivan +". The following is some information about each of these games:

2.1.1 Seeing AI:

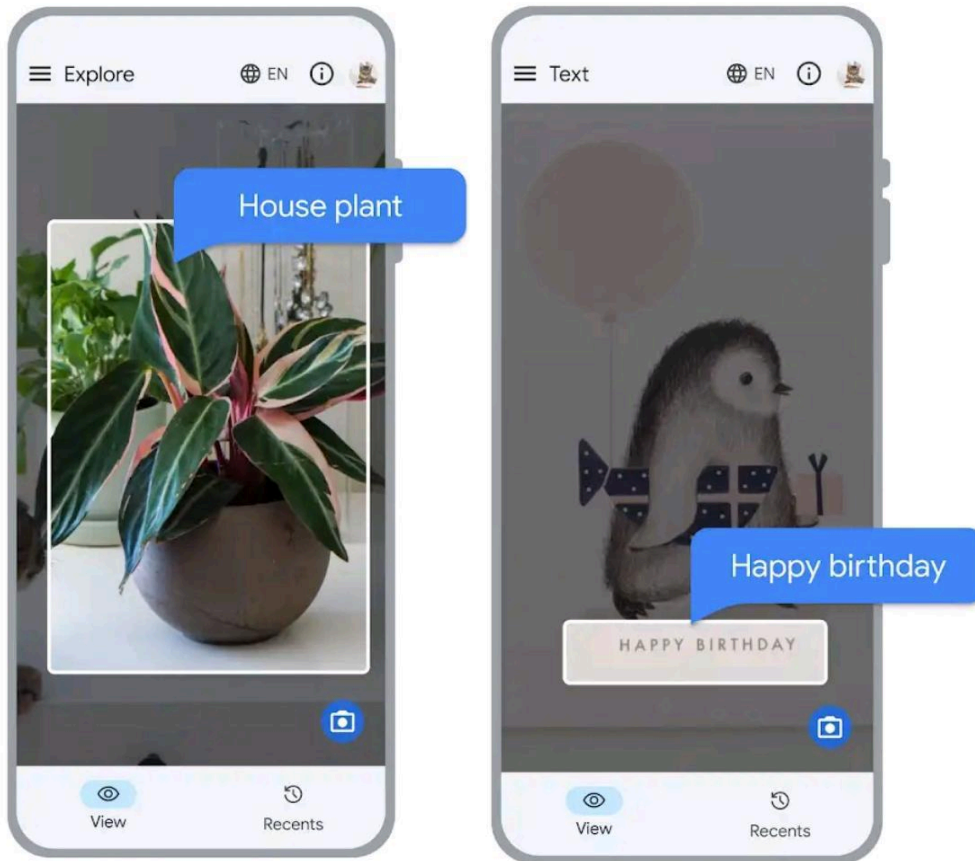
Microsoft's Seeing AI is a free mobile application designed to assist individuals who are blind or have low vision by narrating the world around them using artificial intelligence. Initially launched for iOS devices in 2017, the app became available on Android devices on December 3, 2024, coinciding with the International Day of Persons with Disabilities.



2.1.2 Look Out:

Google's Lookout app is designed to assist individuals who are blind or have low vision by using computer vision and artificial intelligence to interpret and describe their surroundings through their smartphone's camera. The app is available in more than 30 languages and requires Android 6.0 or higher, with a recommendation of at least 2GB of RAM for optimal performance. Since its initial announcement in 2018, Lookout has undergone significant updates to enhance its functionality and accessibility. In 2020, Google expanded the app's capabilities to include features like food label recognition, allowing users to identify packaged foods by their labels or barcodes.

To use Lookout, users can position their smartphone's camera to capture their surroundings, and the app provides spoken notifications about identified objects, text, or people. This hands-free approach enables users to stay engaged with their activities while receiving real-time information about their environment.

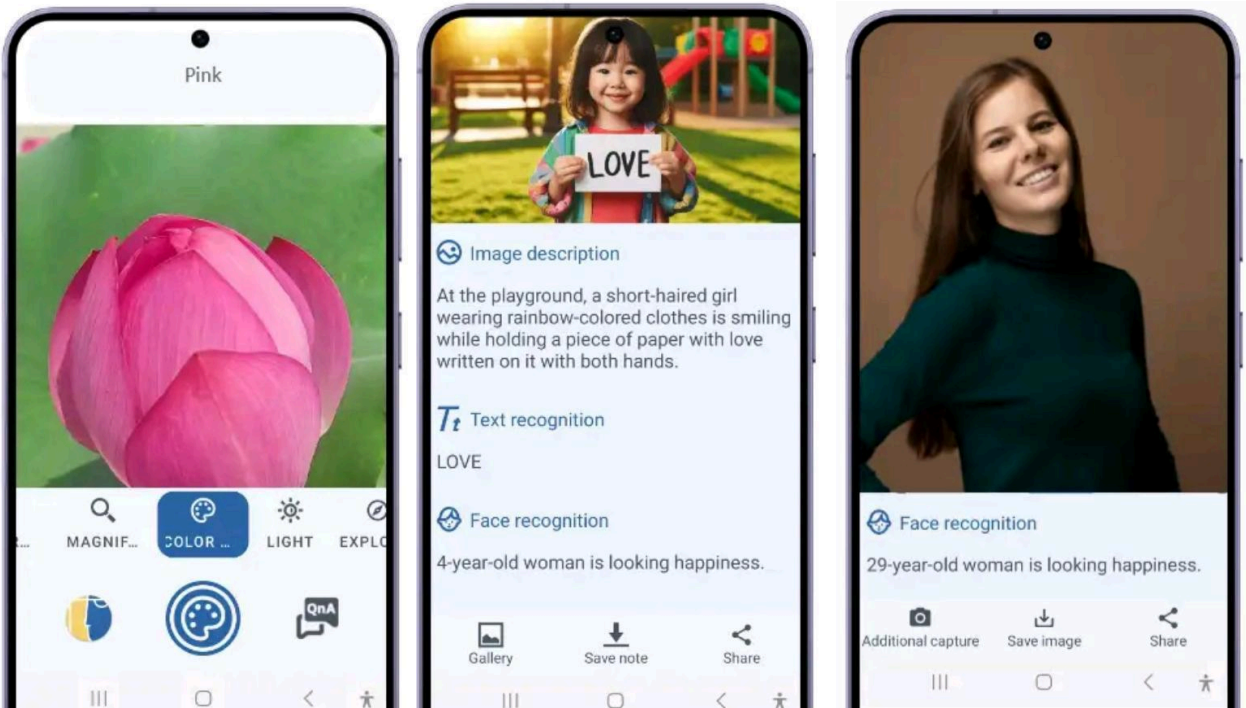


2.1.3 Sullivan +:

Sullivan Plus is a mobile application developed by the South Korean company TUAT, designed to assist visually impaired and elderly users by providing information about their surroundings through smartphone cameras. Launched on November 19, 2019, the app utilizes artificial intelligence to offer voice guidance, helping users navigate city streets and identify objects or text in their environment.

Sullivan Plus supports multiple languages, including Korean, Chinese, English, Japanese, Russian, and Spanish, making it accessible to a broad user base. The app is available for download on both Google Play and the App Store, enabling users across different platforms to benefit from its features.

In December 2019, LG Uplus collaborated with TUAT to enhance the app's capabilities, adding more languages and rolling out an international version for Android users outside Korea. An iOS version was also introduced, expanding its accessibility.



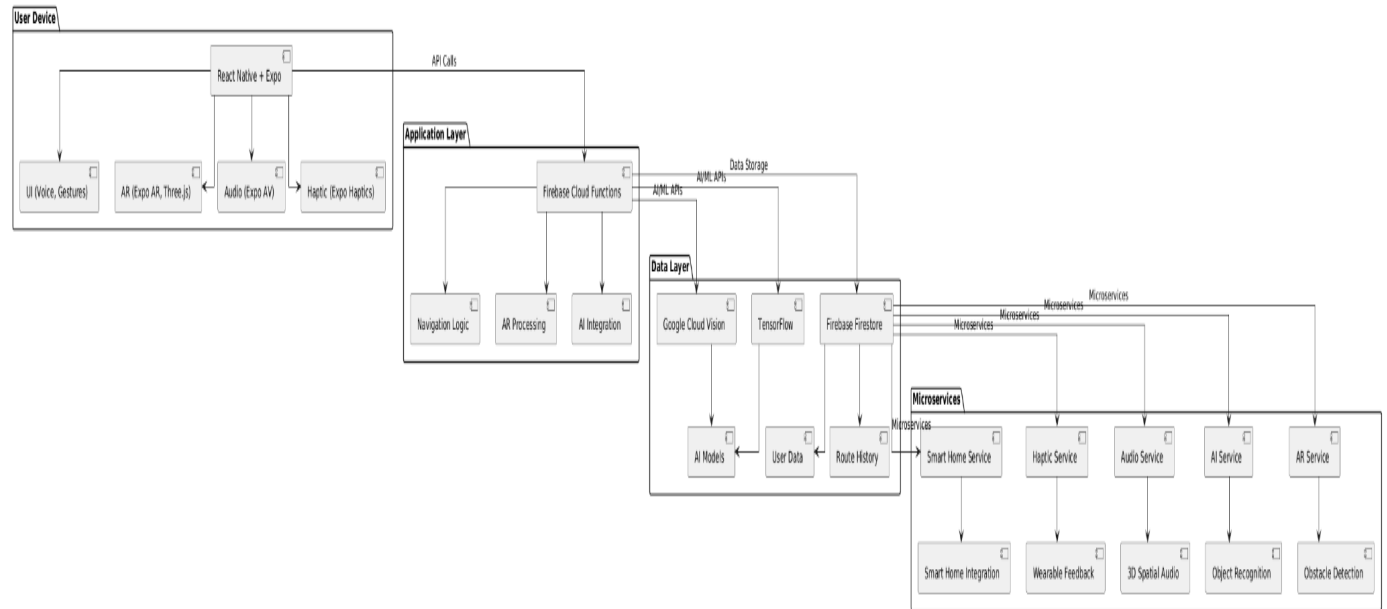
3 Proposed Methodology and Architecture:

3.1 Methodology:

The development of Sensei will follow an Agile methodology, for its flexibility, iterative progress, and continuous user feedback. Sensei is designed for visually impaired users, so their feedback and needs must be at the center of development. Agile has continuous user involvement through regular feedback. This makes sure that the app meets real-world needs of blind users. Also, building an AR app with AI includes many evolving requirements. But through Agile, the team can adapt to changes quickly and easily. If a feature isn't working or a user suggests a better approach, the team can change directions without derailing the entire project.

3.2 Architecture:

The architectural model for Sensei will follow a Layered Architecture combined with Microservices for specific functionalities.



3.3 Individual Tasks :

Software development of “Sensei” is divided into 5 phases :

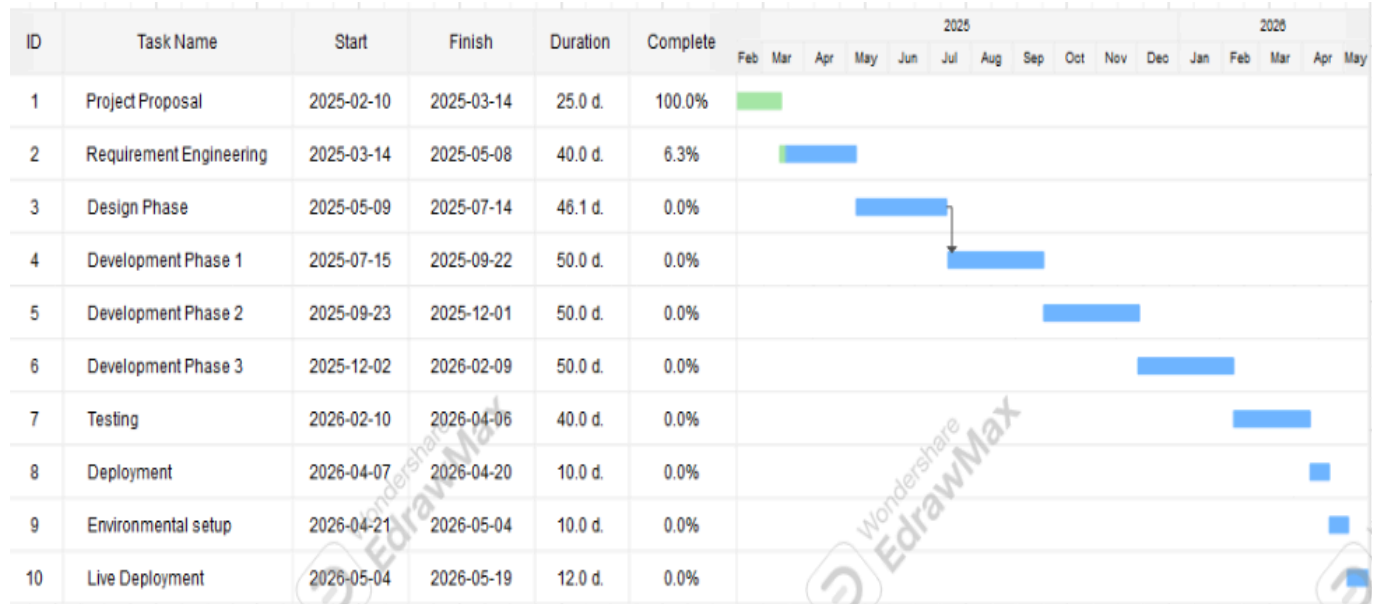
1. **Requirement Analysis** : Gathering Requirements from blind users , accessibility experts and caregivers .
2. **Design** : Designing wireframes and prototypes using Figma while focusing on audio-first design and haptic feedback integration .
3. **Development**: Building the app using AR / AI and cross-platform mobile application development tools .
4. **Testing** : Rigorous testing, including unit, end-to-end, and user testing with visually impaired people to ensure that the app is functional , accessible and user-friendly .
5. **Deployment** : Releasing the app to the public and ensuring smooth on-boarding.

Team Member	Activity	Tentative Date
Jaisha,Neha	Project planning	14/03/25

Team Member	Activity	Tentative Date
Jaisha	Requirement Gathering	14/04/25
Neha	Develop Functional Requirement	14/05/25
Neha	Develop non-functional requirement	22/05/25
Jaisha, Neha	UML diagrams	06/06/25
Neha	UI/UX	20/09/25
Neha	Front-end Development	20/10/25
Jaisha	Back-end Development	20/11/25
Jaisha	Database and architecture	1/12/25
Jaisha	Integration	1/04/26
Neha	Testing	16/04/26
Neha,Jaisha	Deployment	14/05/26

3.4 Gantt Chart:

The following gantt chart graphically represents the tasks mentioned above.



4 References:

1. “ Look Out” :

Official Blog: Google Blog - Lookout App

BBC Article: [BBC - Google’s Lookout Update](#)

2. “ Seeing AI ”:

Official Blog: [Microsoft Blog - Seeing AI](#)

3. “ Sullivan+ ” :

Official Website: [mysullivan.org](#)

Article: PR Newswire - Sullivan+ AI-Based App

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